



Environment and
Climate Change Canada

Environnement et
Changement climatique Canada

CWS PATHWAY OF EFFECTS SHINY APP TOOL

User manual for installation and
operation

July 2026 v2



Canada 



QUICK REFERENCE

- If you are opening this manual for the first time, please read through all the slides and follow the instructions for software and file download and app set-up.
- If you have already installed the necessary software and R packages and are re-visiting this manual for guidance on how to re-open the app, please skip ahead to [**Step 4**](#) [**on Slide 42**](#)
- Advanced users can view the project code and install from the source code on GitHub [here](#).



OVERVIEW

- What is the Pathway of Effects (PoE) Shiny App tool?
 - An application developed to allow practitioners to build automated and custom PoE models, which can then be used by practitioners to inform project assessments
 - The app adds graphically represented pathways in a flowchart based on selected activities for a specified Valued Component (VC), based on the user's inputs into the program.
 - The present version must be operated using R or RStudio. No file downloads are necessary.



OVERVIEW

- Purpose of this Guide: To show Impact Assessment practitioners how to operate the PoE Shiny App Tool
- Contents:
 - Slide 5-17: [Step 1 – How to install R on your PC](#)
 - Slide 18-28: [Step 2 – How to install RTools](#)
 - Slide 29-34: [Step 3 – How to install RStudio](#)
 - Slide 35-41: [Step 4 – How to install and launch PoE tool](#)
 - Slide 42: [Step 5 – Quick Launch \(after first installation\)](#)
 - Slide 43-51: [Step 6 – How to open and use the Shiny App](#)
- Notes:
 - First-time setup is estimated to take 20-40 minutes.
 - Steps 1-3 only need to be completed once
 - If you already have both R, RTools, and RStudio installed on your PC, [skip to step 4](#)



STEP 1: INSTALL R ON YOUR PC

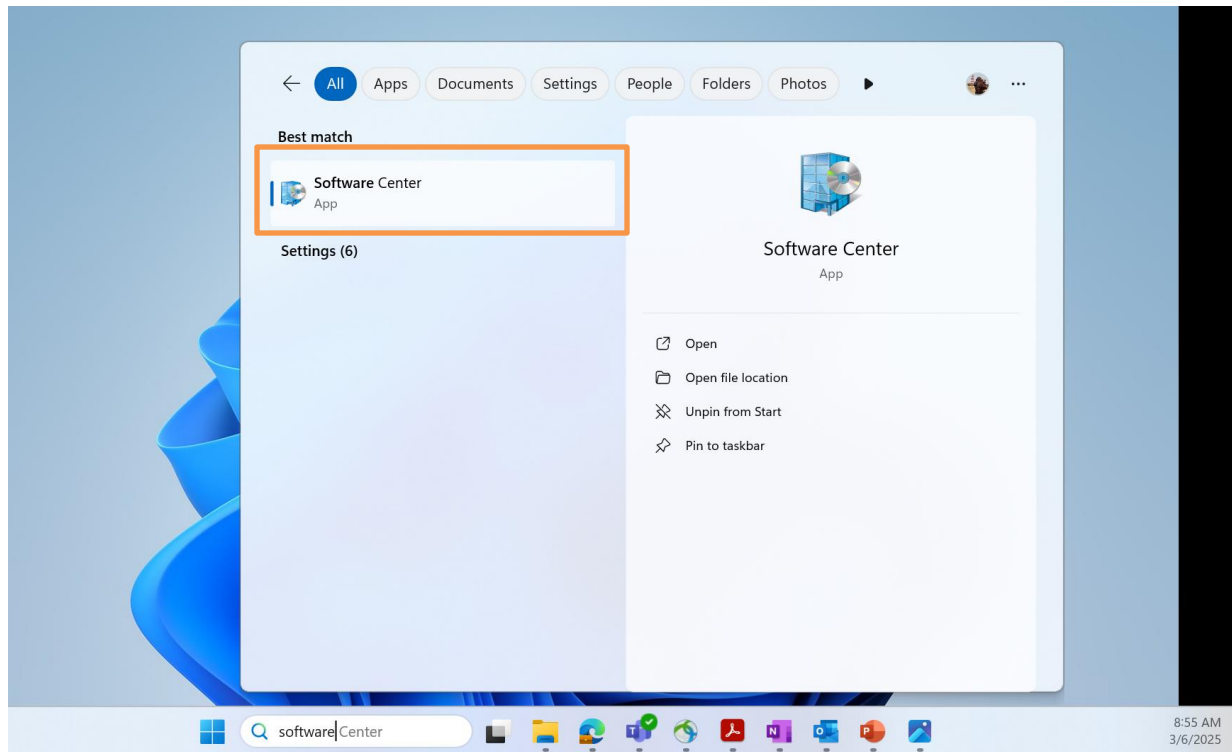
- This step only needs to be completed once. If you already have R installed on your PC, [skip to Step 2 \(slide 18\) to install RTools](#)
- R needs to be installed before RTools.





STEP 1: INSTALL R ON YOUR PC

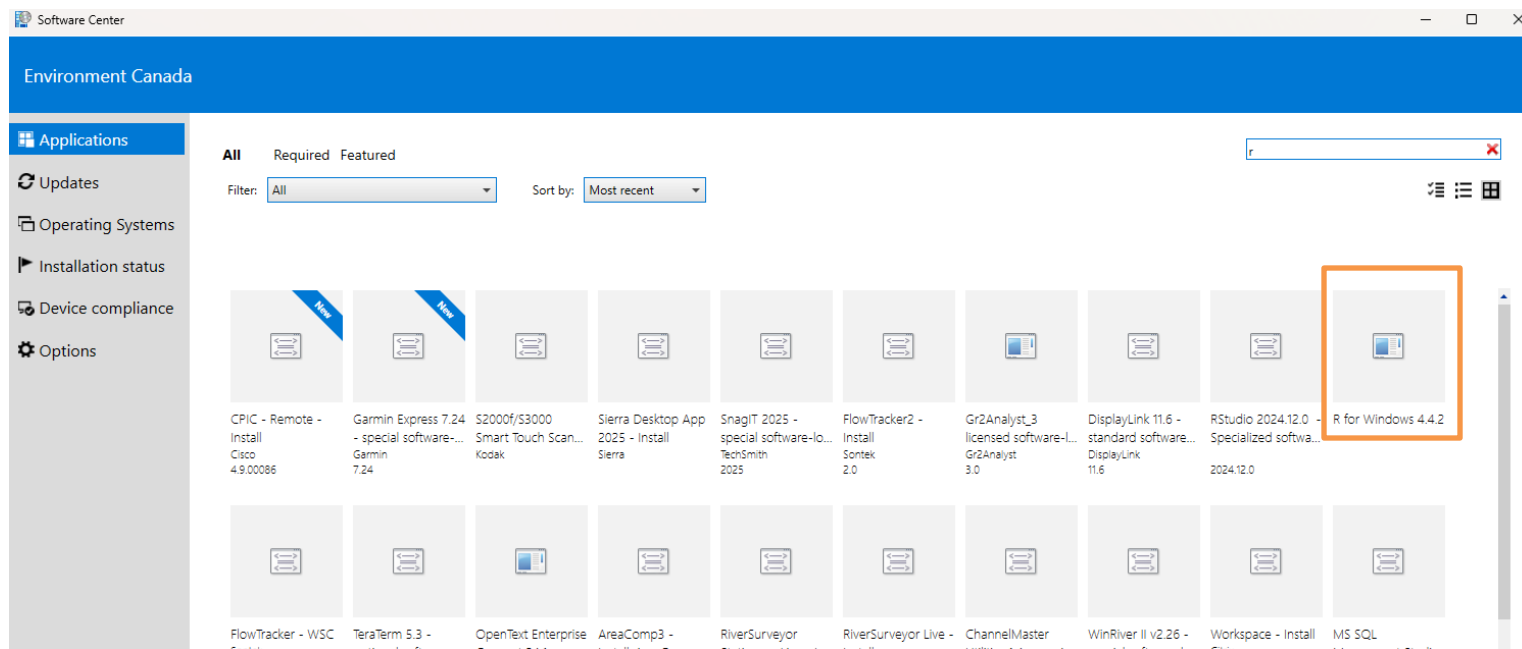
- 1.1 – Find and open the Software Center





STEP 1: INSTALL R ON YOUR PC

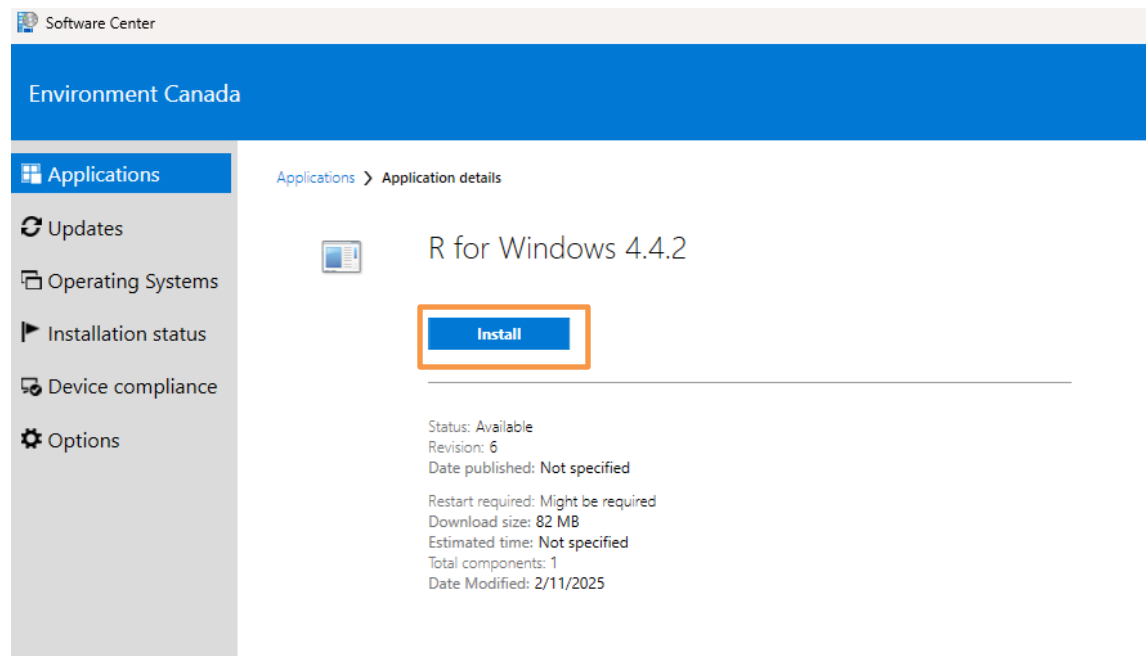
- 1.2 – Find or search “R for Windows”





STEP 1: INSTALL R ON YOUR PC

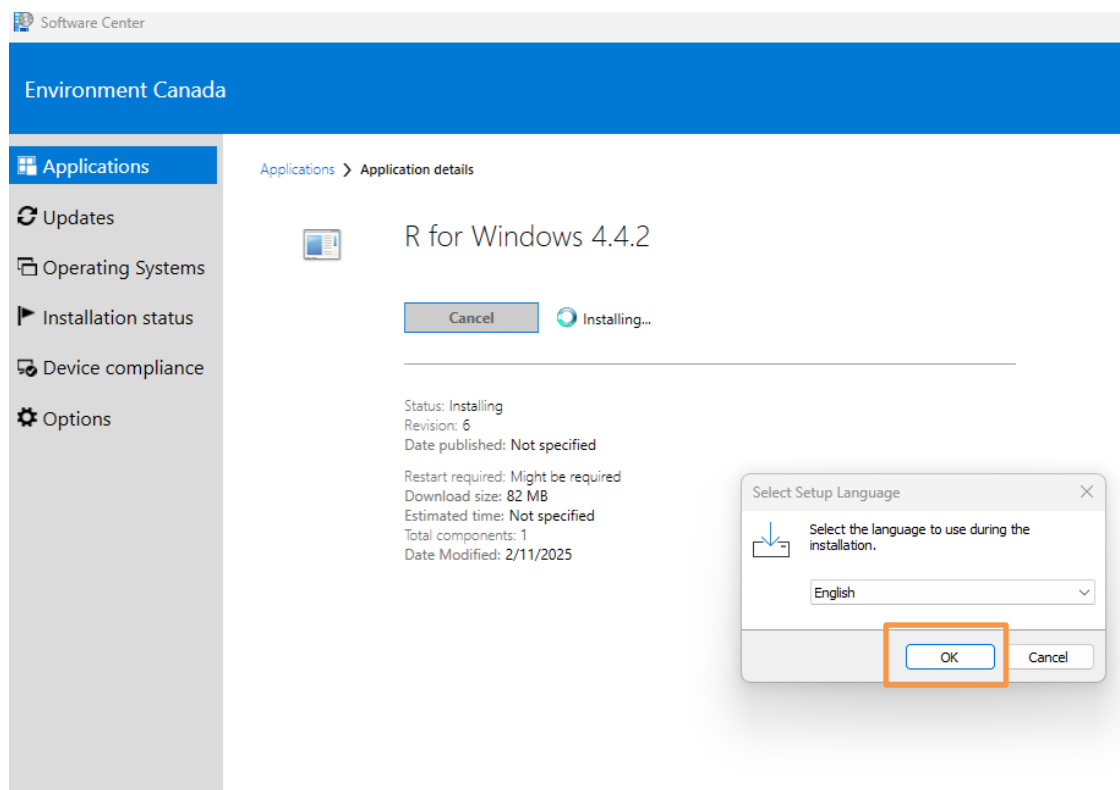
- 1.3 – Click “Install”





STEP 1: INSTALL R ON YOUR PC

- 1.4 – Select preferred installation language

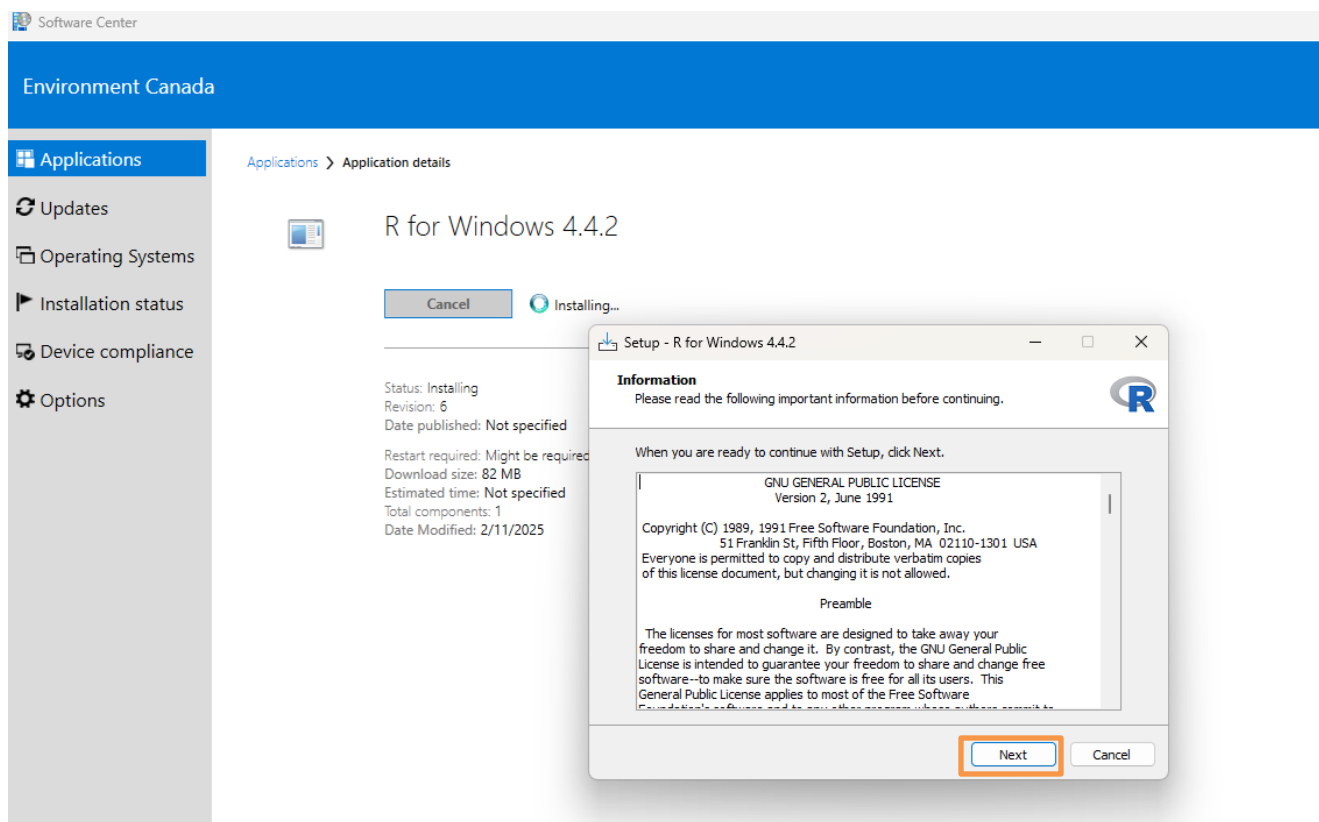




STEP 1: INSTALL R ON YOUR PC

- 1.5 – When “Setup” window pops up, click “Next”

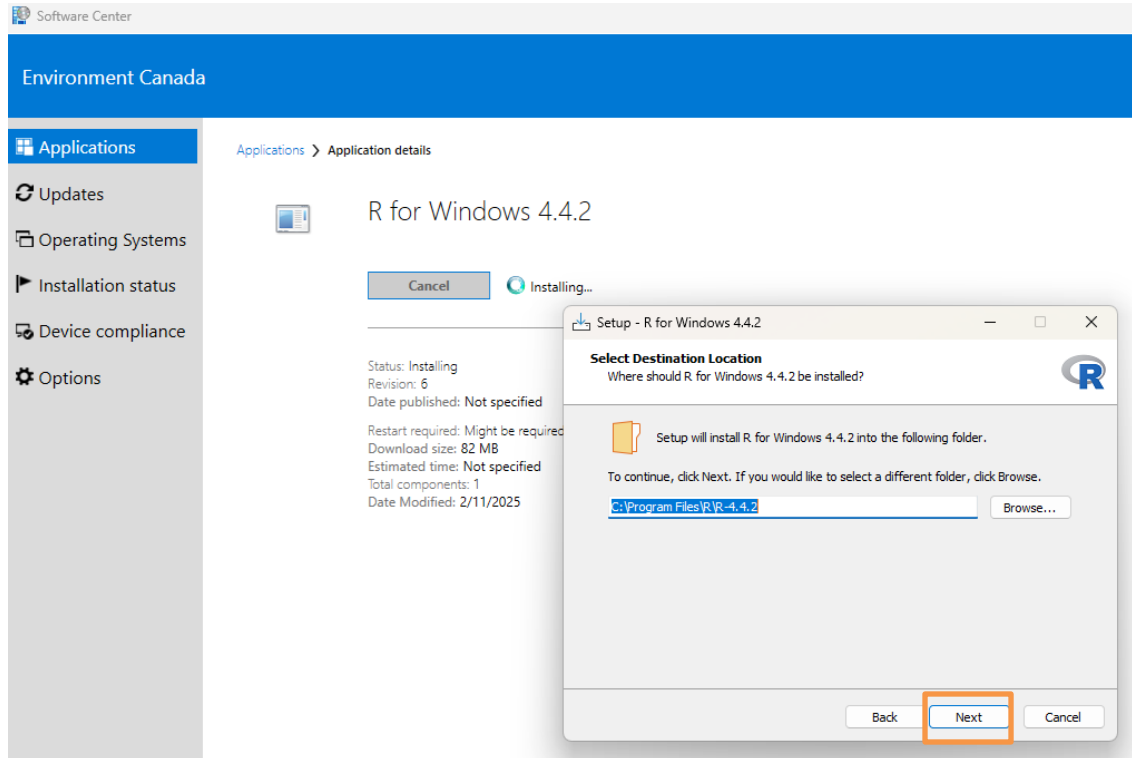
Note: It may take a few minutes or more before this window pops up.





STEP 1: INSTALL R ON YOUR PC

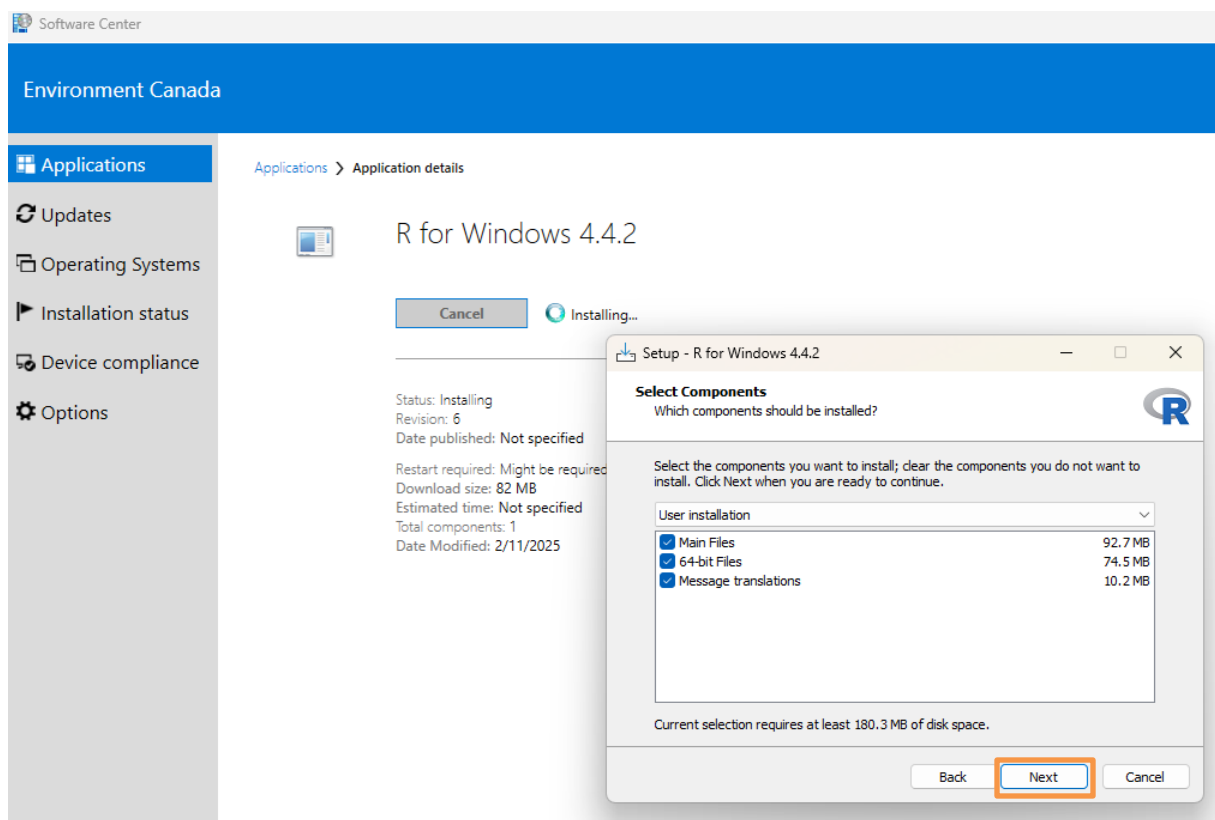
- **1.6** – Click “Next” for default installation location or browse to select other preferred location on your PC





STEP 1: INSTALL R ON YOUR PC

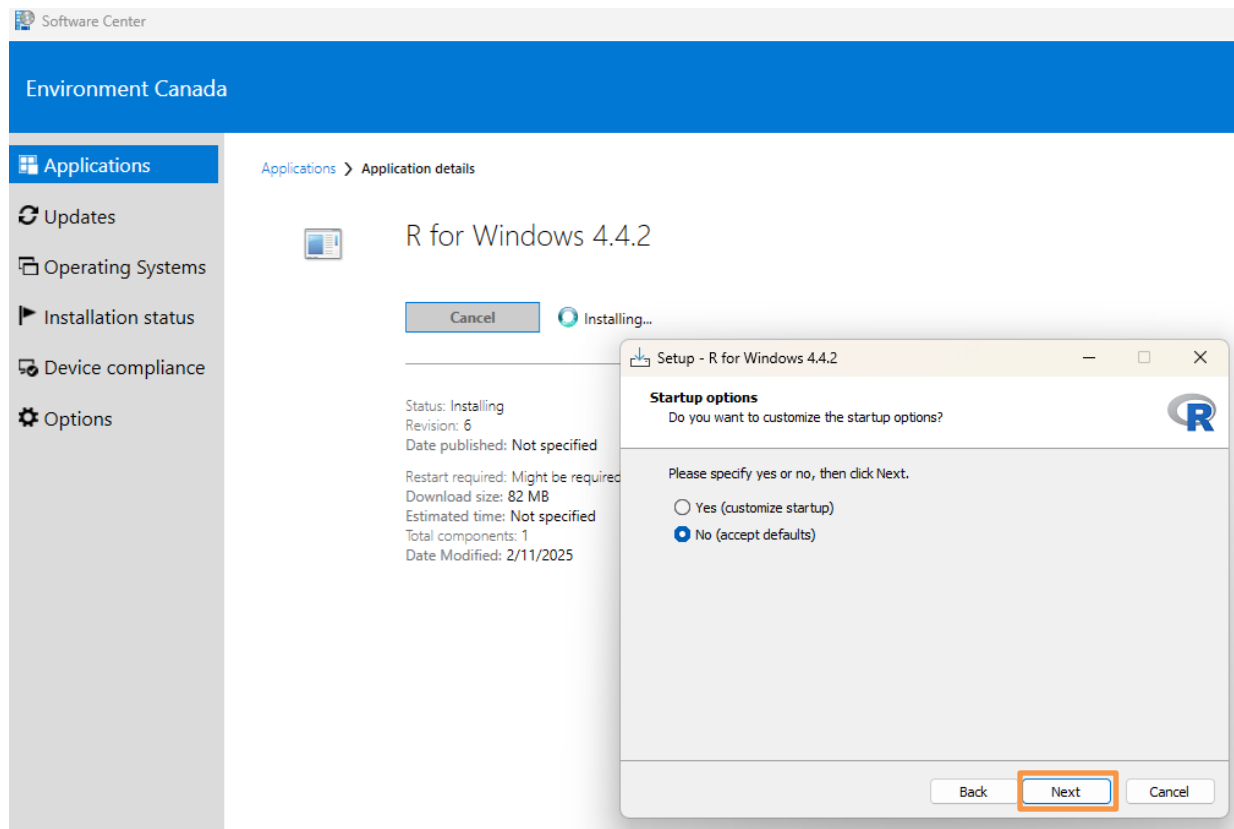
- 1.7 – Click “Next” to accept default components





STEP 1: INSTALL R ON YOUR PC

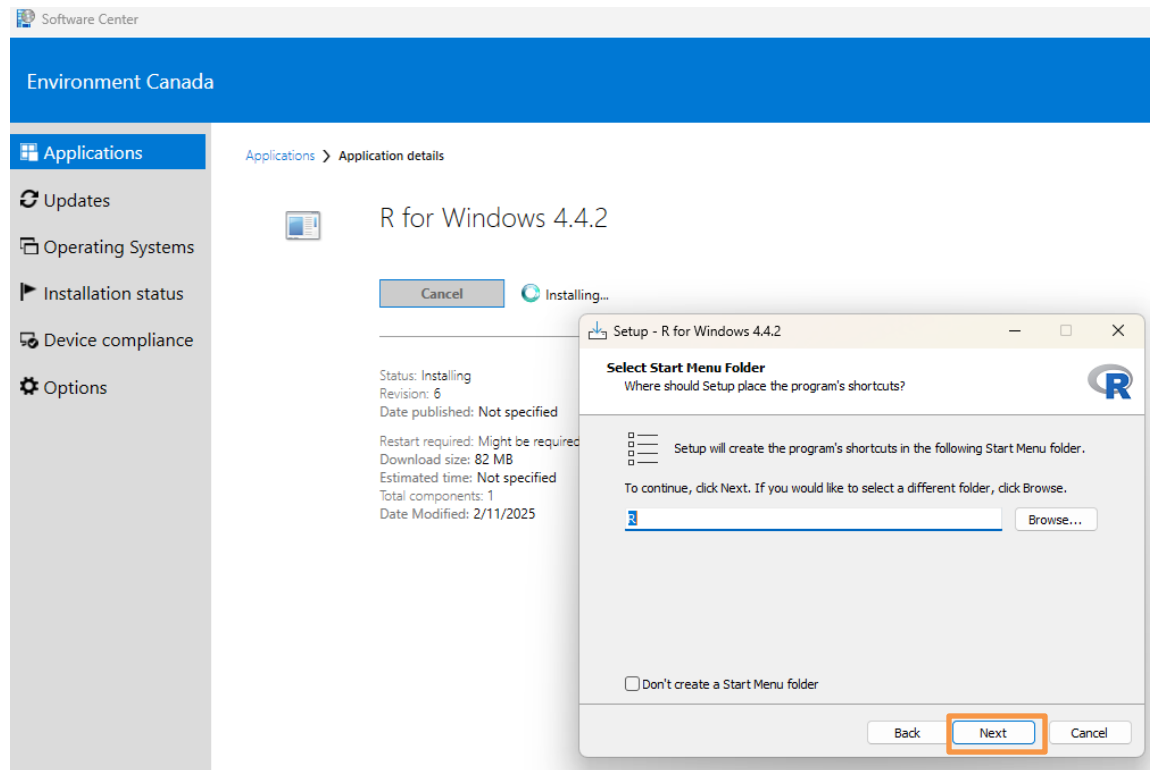
- 1.8 – Click “Next” to accept default startup options





STEP 1: INSTALL R ON YOUR PC

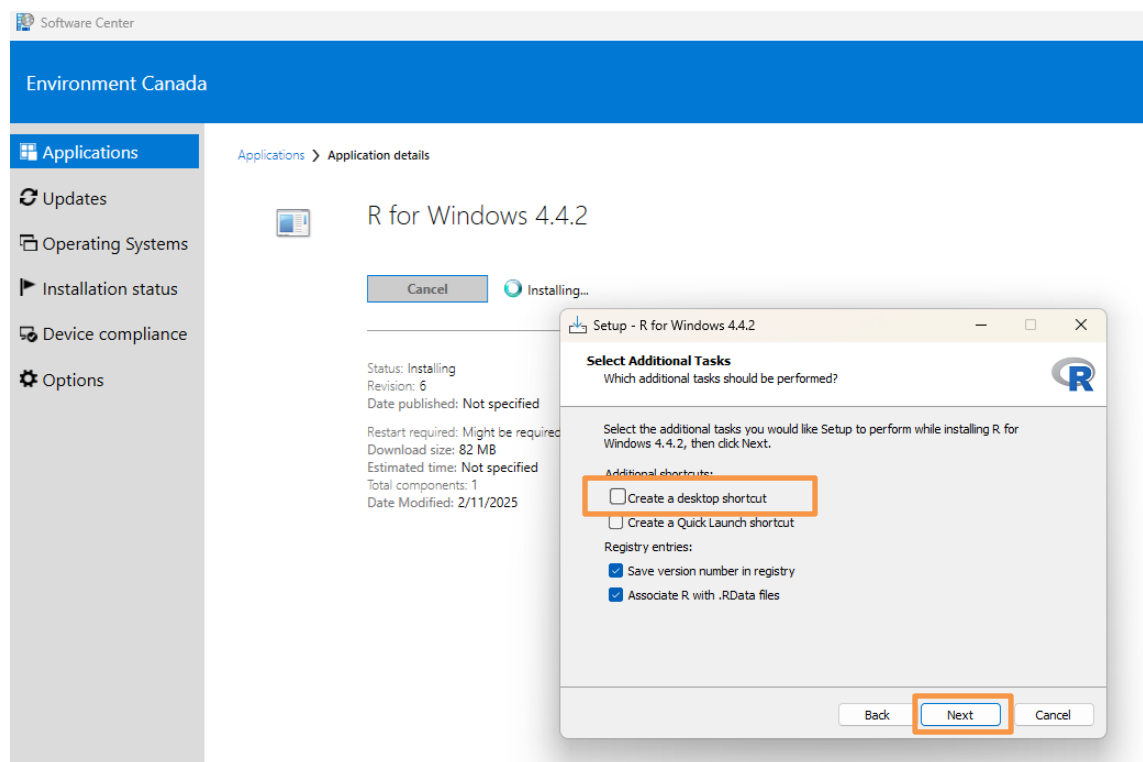
- **1.9** – Click “Next” to select the default Start Menu folder location or browse to other preferred location on your PC





STEP 1: INSTALL R ON YOUR PC

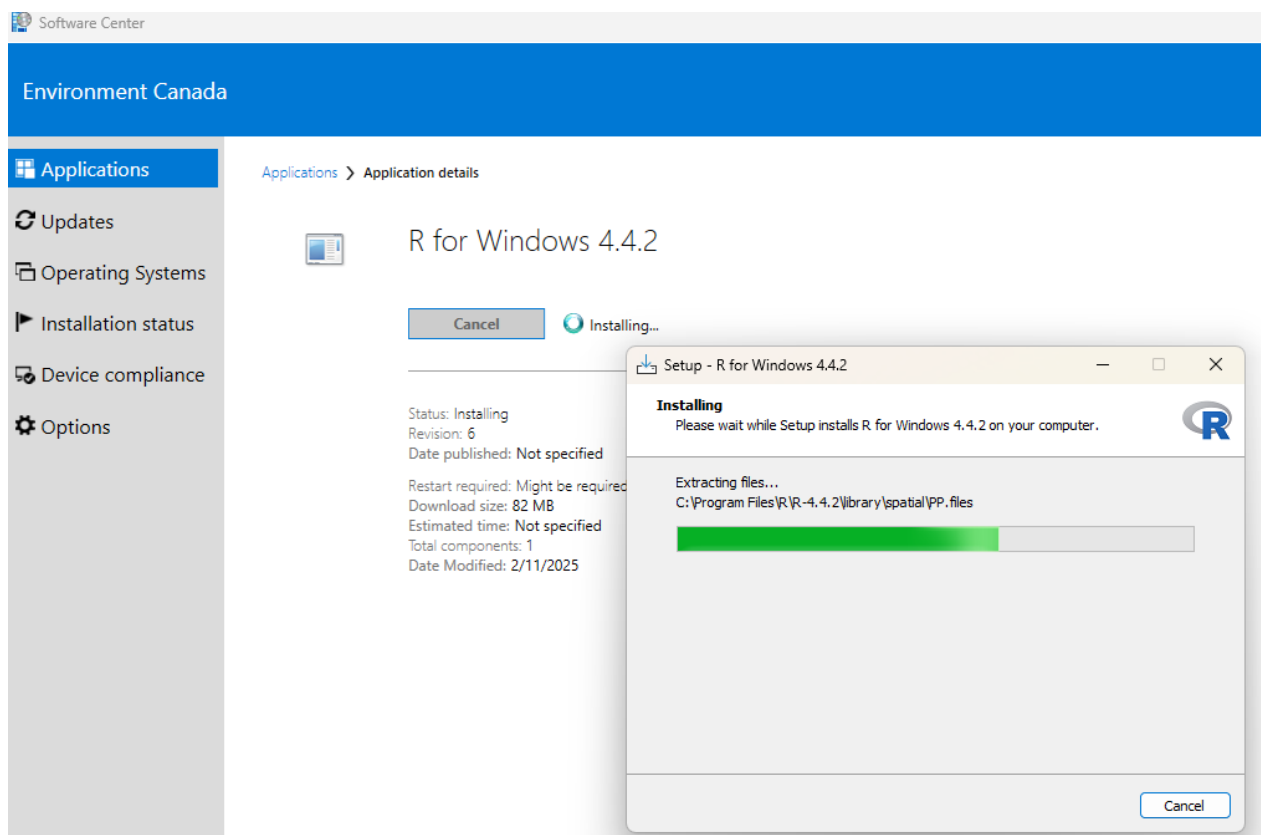
- **1.10** – Uncheck the box “Create a desktop shortcut” then click “Next”





STEP 1: INSTALL R ON YOUR PC

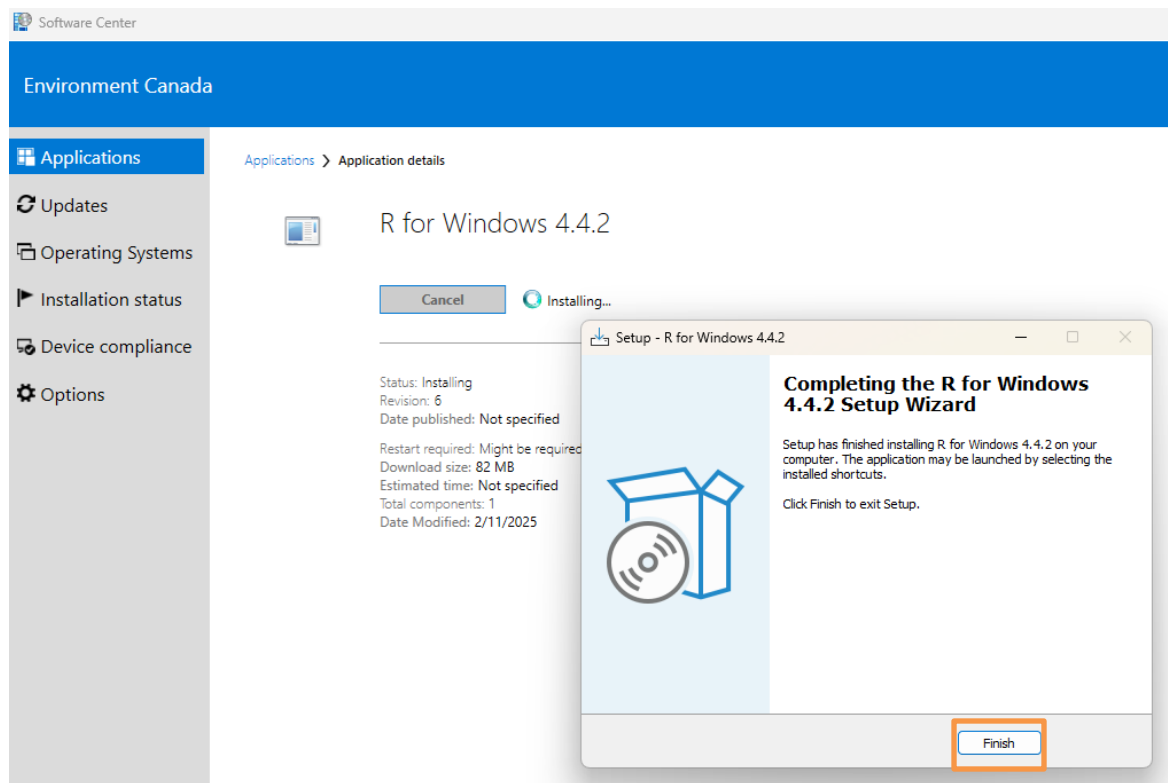
- 1.11 – Wait for program to install





STEP 1: INSTALL R ON YOUR PC

- 1.12 – Click “Finish”





STEP 2: INSTALL RTOOLS

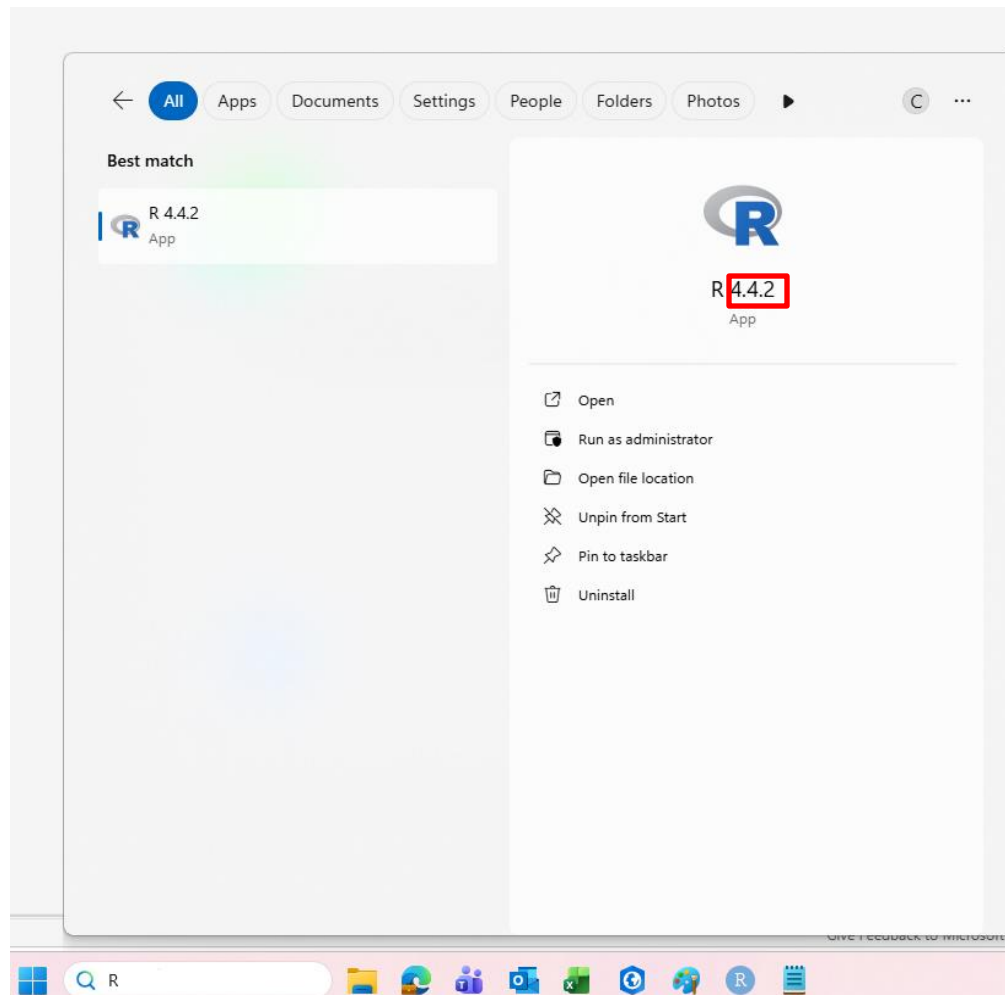
- To install the PoE Shiny App tool, we need to install RTools to be able to "build" the software.
- This step only needs to be completed once. If you already have RTools installed on your PC, skip to [Step 4 \(slide 35\)](#) to install the PoE Shiny App Tool.





STEP 2: INSTALL RTOOLS

- **2.1** – Find the R version number by searching "R" in the search bar at the bottom of the screen.
- Note the version number (here, it is 4.4.2)





STEP 2: INSTALL RTOOLS

- **2.2** – Go to the RTools download page

[RTools: Toolchains for building R and R packages from source on Windows](#) (Ctrl + Click to go to this link)





STEP 2: INSTALL RTOOLS

- **2.3** – On the RTools download page, select the version of RTools with the version number that most closely matches your R's version number.
- E.g. If R's version number is 4.4.2, select RTools version 4.4. If R's version number is 4.5.1, select RTools 4.5.

Choose your version of Rtools:

[RTools 4.5](#)

[RTools 4.4](#)

[RTools 4.3](#)

[RTools 4.2](#)

[RTools 4.0](#)

[old versions of RTools](#)

for R versions from 4.5.0 (including R-devel)

for R versions 4.4.x (R-release)

for R versions 4.3.x (R-oldrelease)

for R versions 4.2.x

for R from version 4.0.0 to 4.1.3

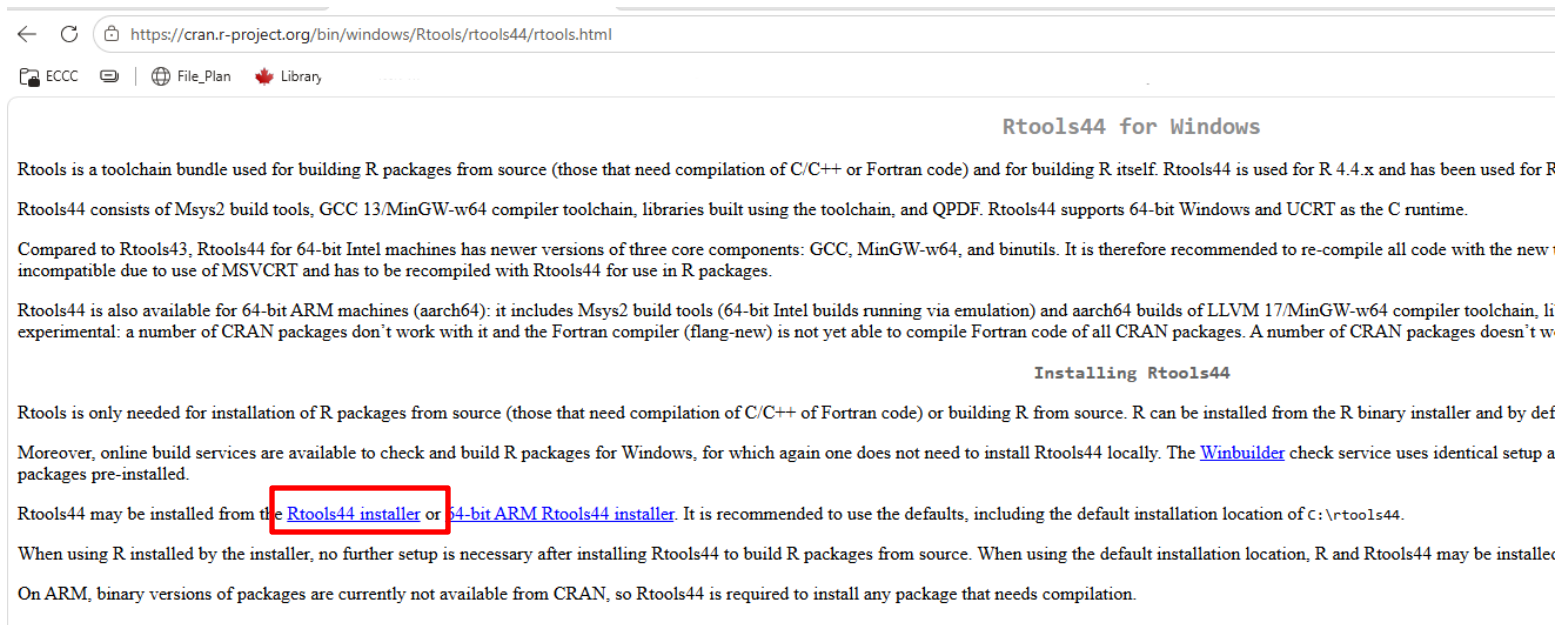
for R versions prior to 4.0.0





STEP 2: INSTALL RTOOLS

- **2.4** – Click the blue Rtools44 installer link (or Rtools45 installer link, depending on your R version number)



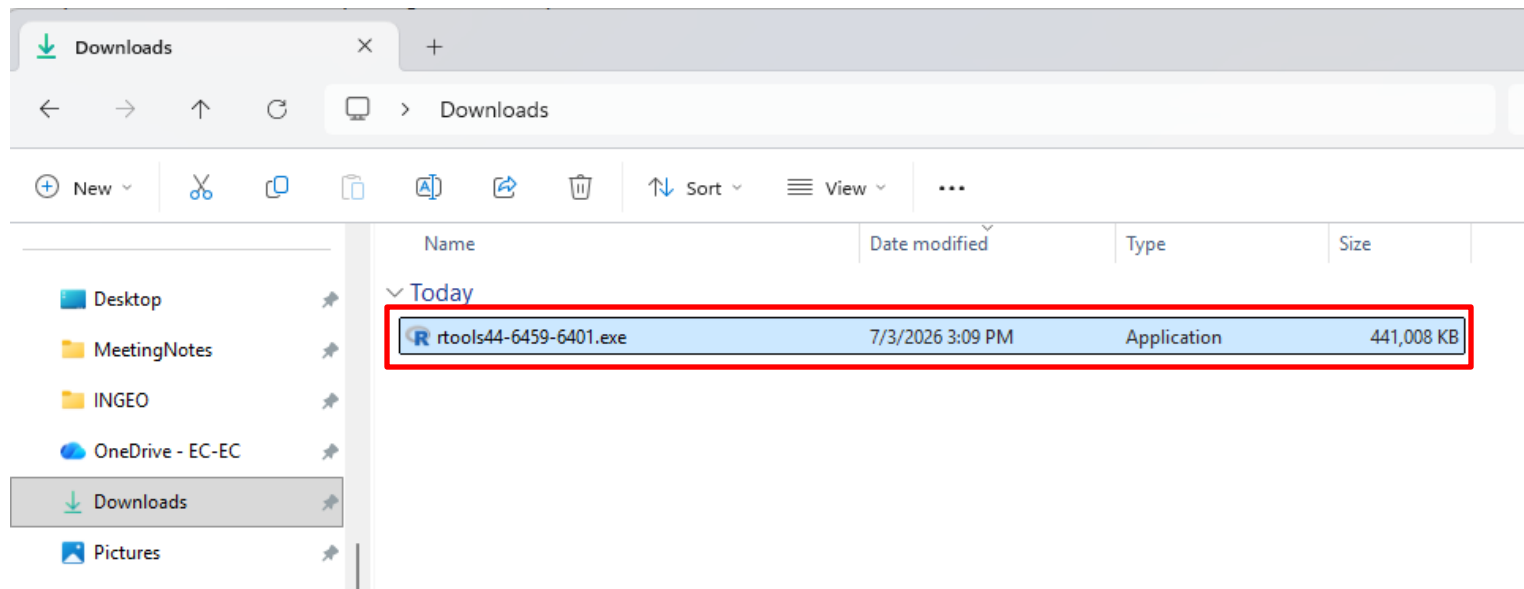
The screenshot shows a web browser window with the address bar displaying `https://cran.r-project.org/bin/windows/Rtools/rtools44/rtools.html`. The page title is "Rtools44 for Windows". The main content describes Rtools as a toolchain bundle for building R packages from source. It mentions that Rtools44 is used for R 4.4.x and consists of Msys2 build tools, GCC 13/MinGW-w64 compiler toolchain, and QPDF. It also notes that Rtools44 supports 64-bit Windows and UCRT as the C runtime. The page further explains that Rtools44 is also available for 64-bit ARM machines (aarch64) and includes Msys2 build tools and LLVM 17/MinGW-w64 compiler toolchain. The "Installing Rtools44" section states that Rtools is only needed for installation of R packages from source or building R from source. It also mentions that online build services are available to check and build R packages for Windows. The text "Rtools44 may be installed from the [Rtools44 installer](#) or [64-bit ARM Rtools44 installer](#)" is highlighted with a red box. The page concludes by stating that when using R installed by the installer, no further setup is necessary after installing Rtools44 to build R packages from source. On ARM, binary versions of packages are currently not available from CRAN, so Rtools44 is required to install any package that needs compilation.





STEP 2: INSTALL RTOOLS

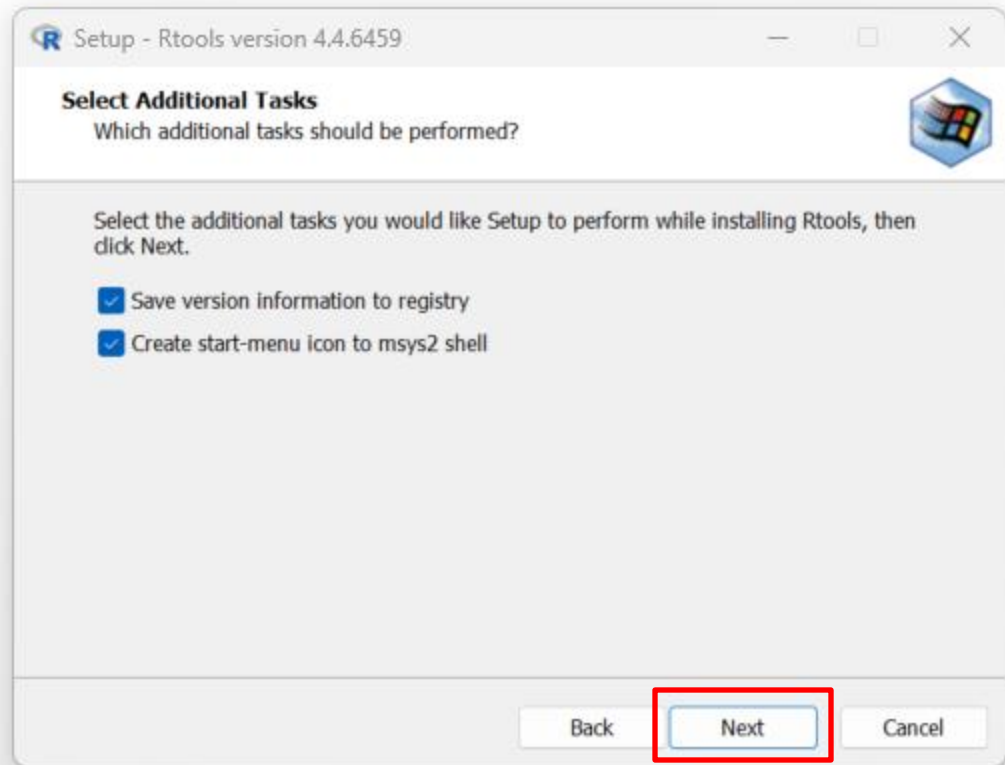
- **2.5** – Double click the RTools executable file you just downloaded.





STEP 2: INSTALL RTOOLS

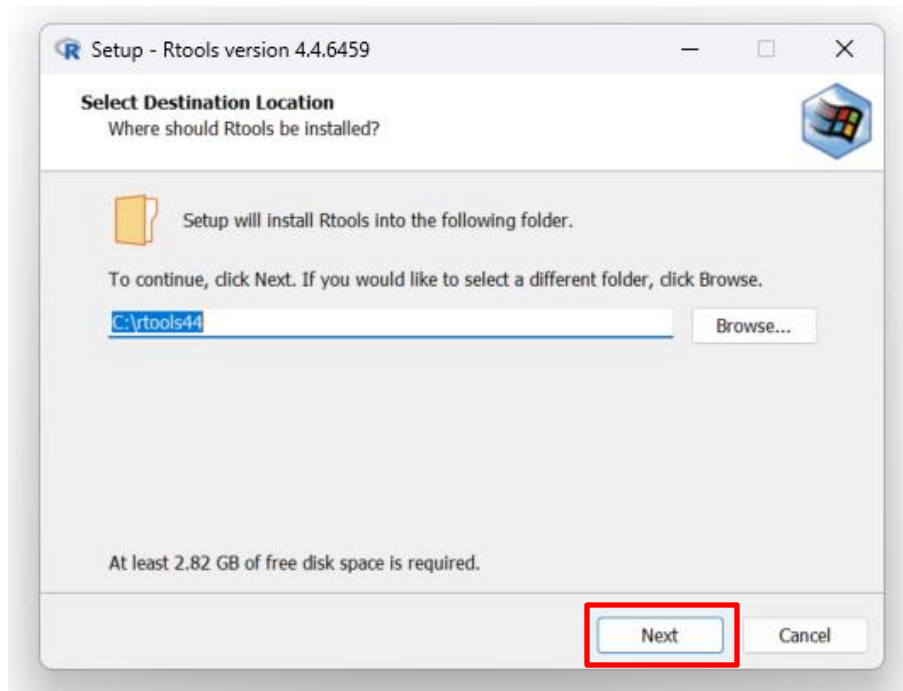
- 2.7 – Click "Next"





STEP 2: INSTALL RTOOLS

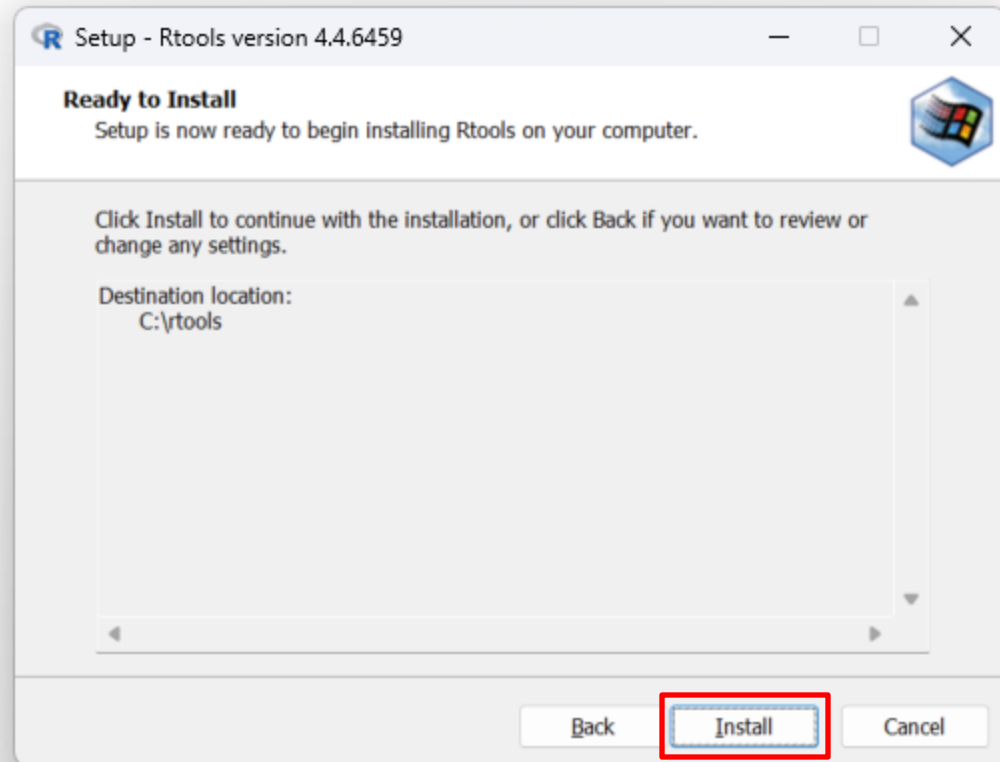
- 2.6 – Click "Next"





STEP 2: INSTALL RTOOLS

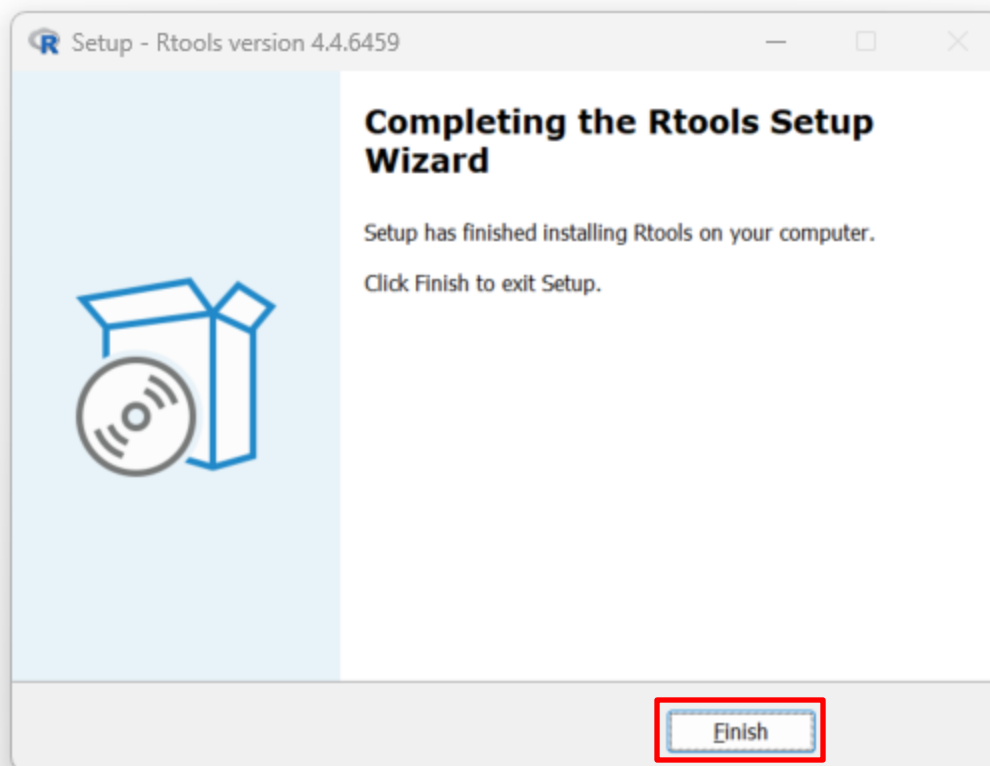
- 2.8 – Click "Install"





STEP 2: INSTALL RTOOLS

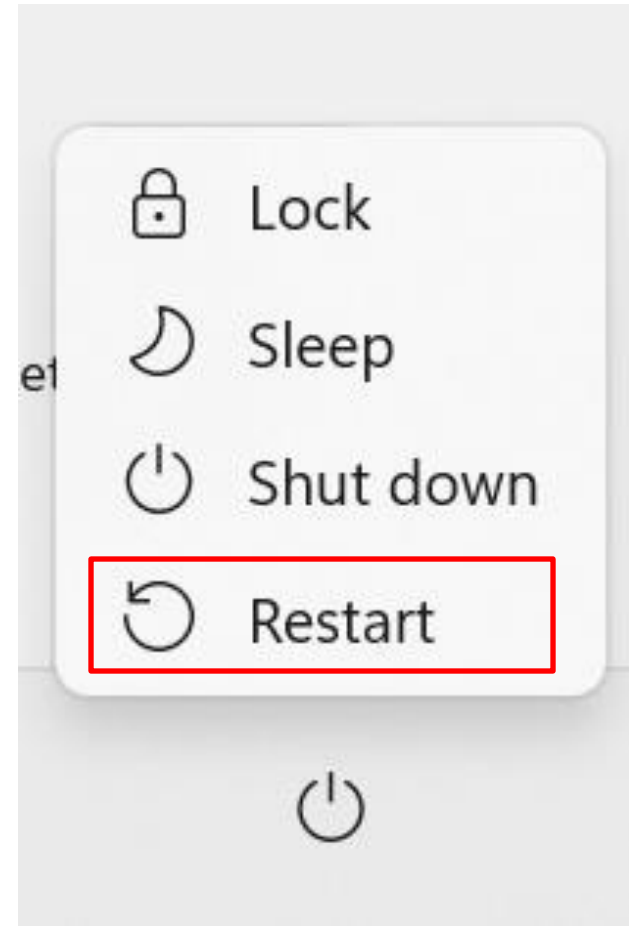
- 2.9 – Wait for installation to complete, then click "Finish"





STEP 2: INSTALL RTOOLS

- 2.9 – Restart your computer.





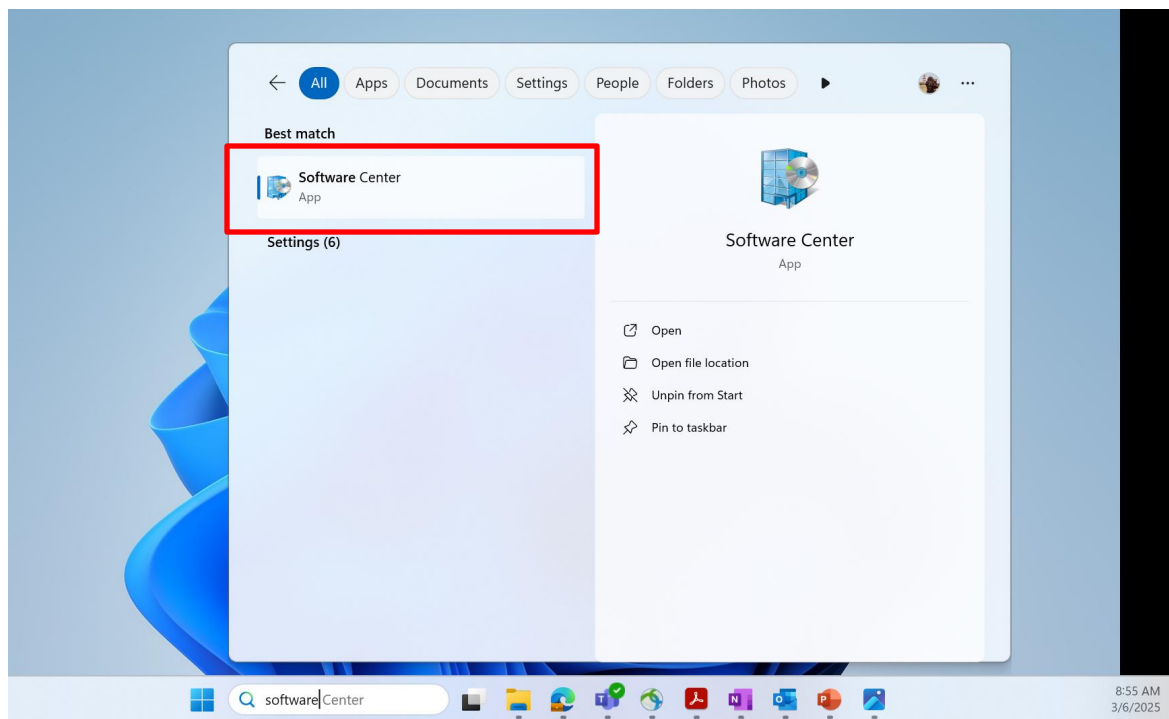
STEP 3: INSTALL RSTUDIO

- RStudio is an "Integrated Development Environment" or "IDE" for R that has many helpful tools built-in.
- One of these tools is Quarto, which allows the PoE Shiny App to create and export reports based on the paths of effect identified.
- The PoE Shiny App **can** be run in the base R console, **but** it cannot export reports because it doesn't have Quarto integrated (you will get an error).
- Installing and running the PoE Shiny App on RStudio ensures the PoE Shiny App will be fully functional on launch.



STEP 3: INSTALL RSTUDIO

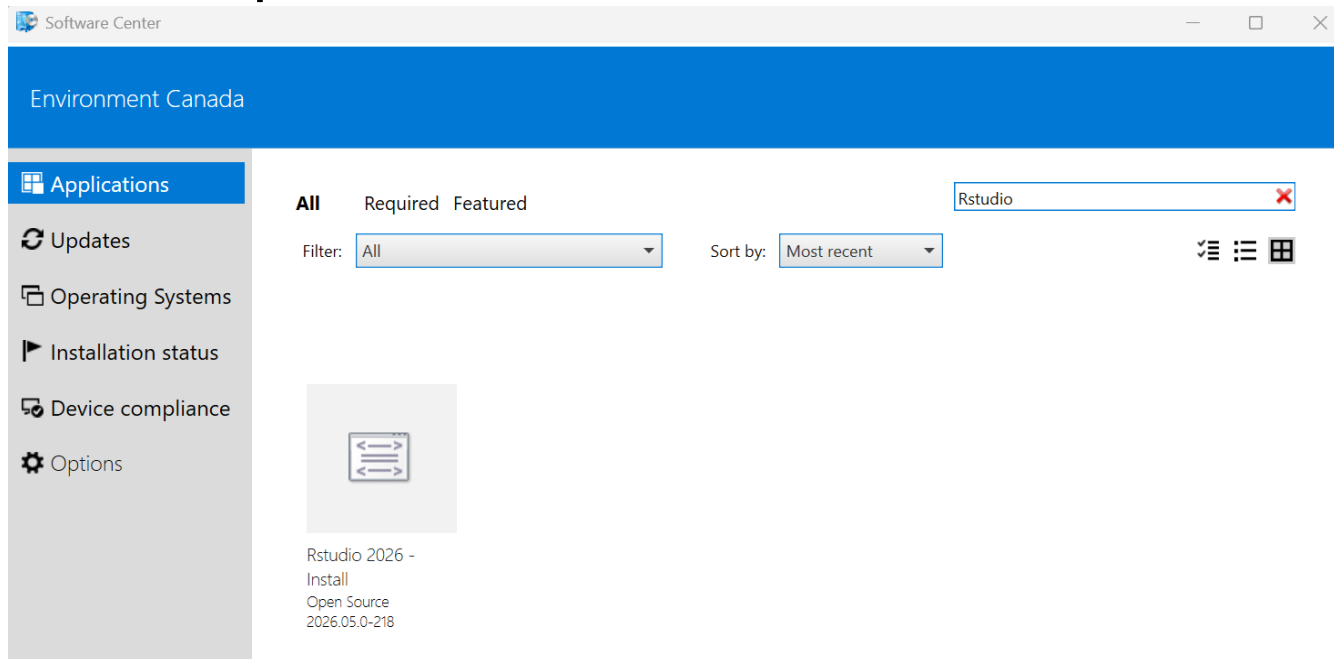
- **3.1** – Find and open the Software Center





STEP 3: INSTALL RSTUDIO

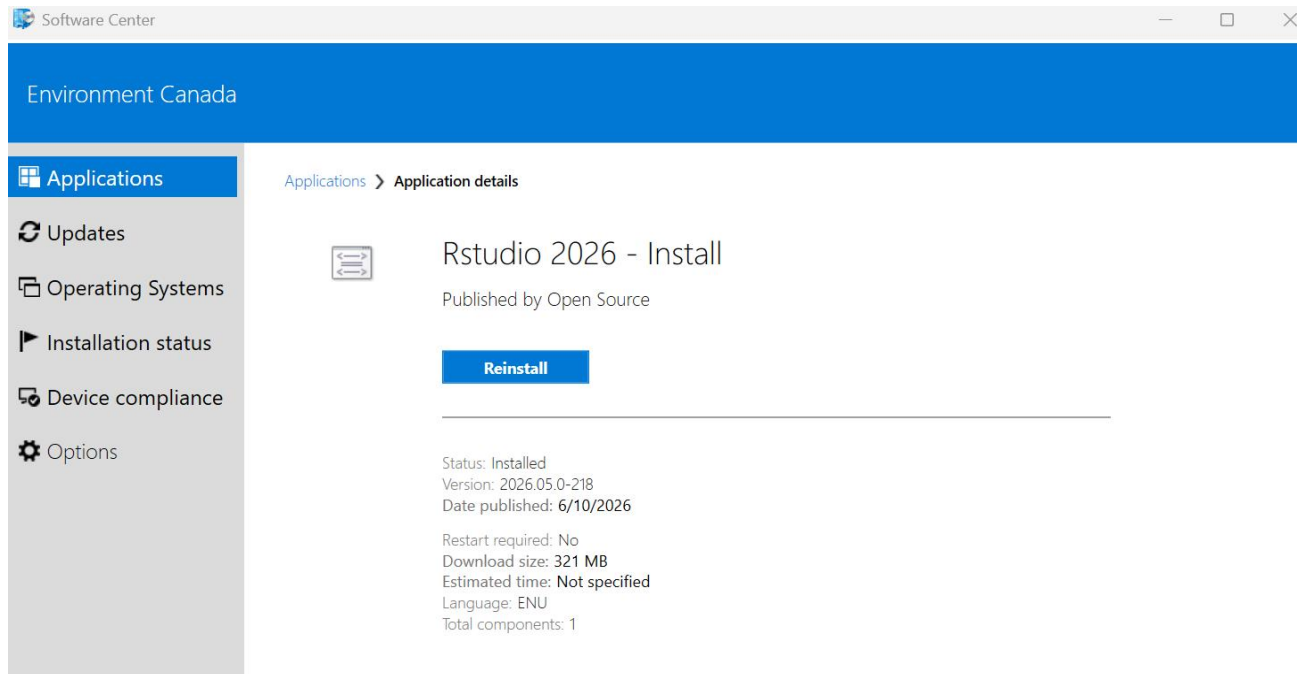
- **3.2** – Enter "RStudio" into the search bar and click the tile that comes up





STEP 3: INSTALL RSTUDIO

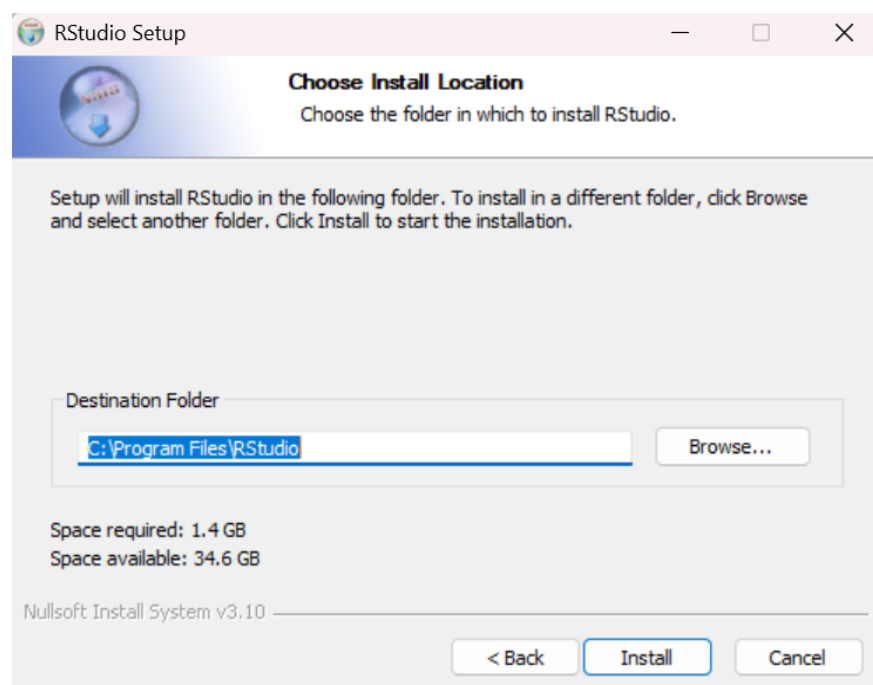
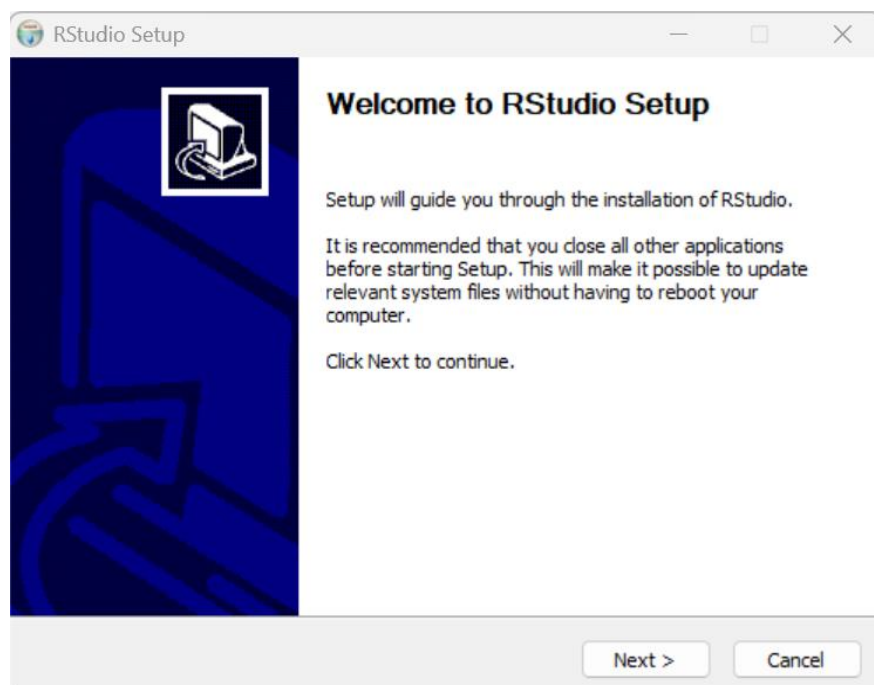
- 3.3 – Click the blue "Install" button





STEP 3: INSTALL RSTUDIO

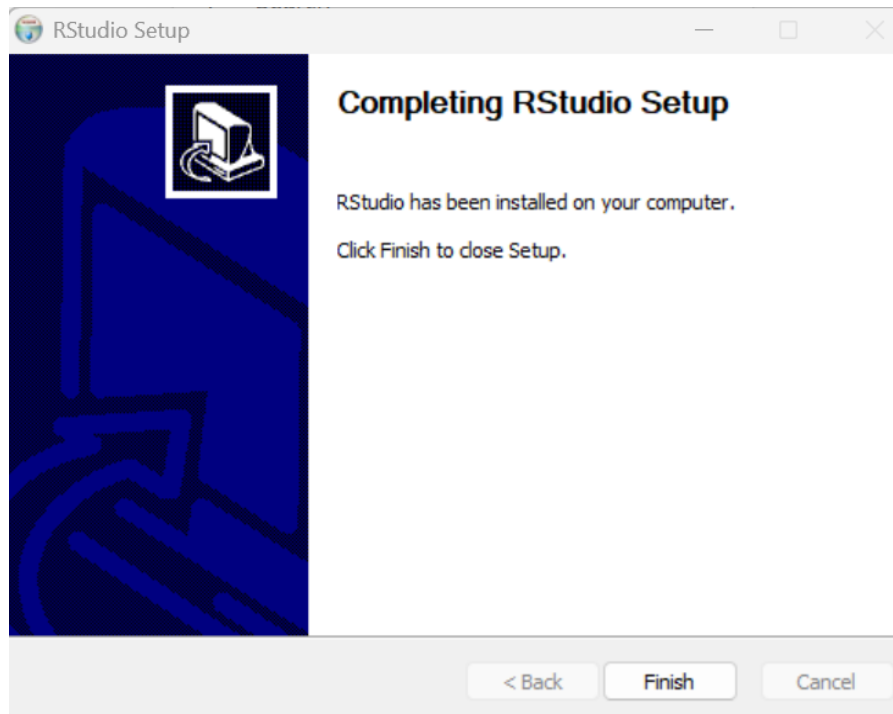
- 3.4 – Click "Next", then click "Install"





STEP 3: INSTALL RSTUDIO

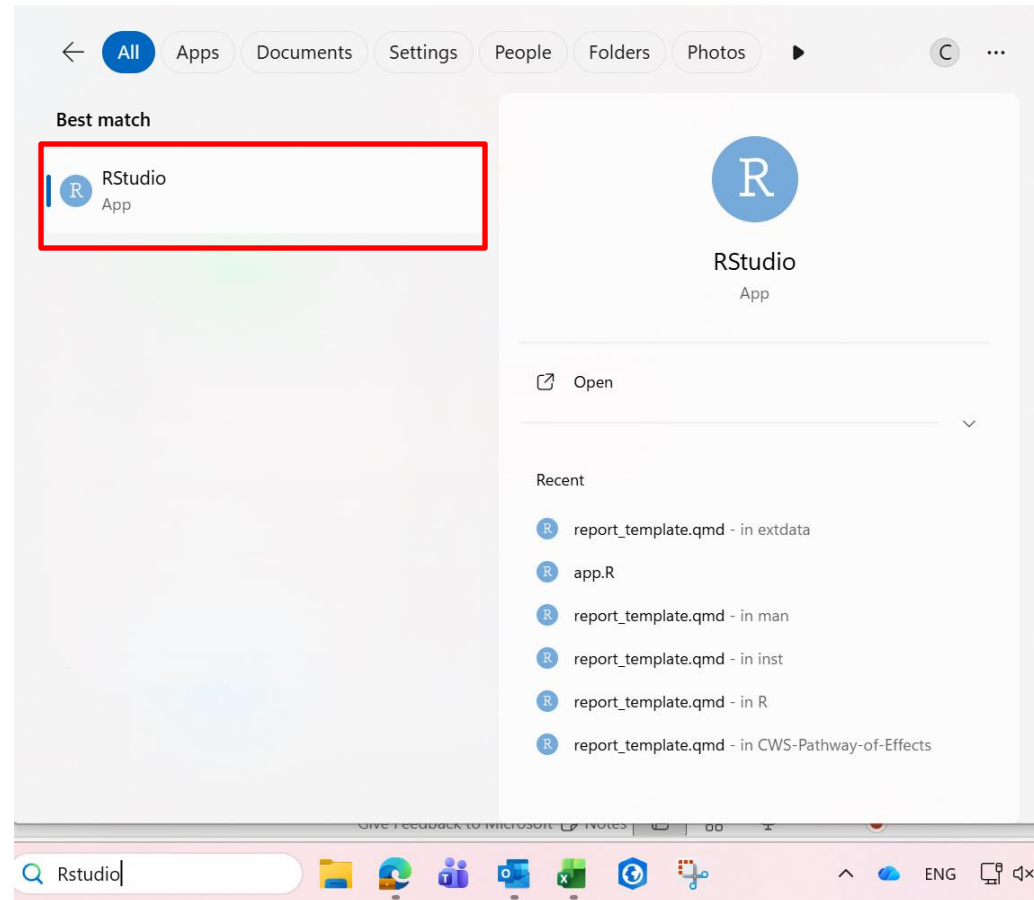
- **3.5** – Wait for RStudio to install, then click "Finish"





STEP 4: INSTALL AND LAUNCH POE TOOL

- **4.1** – After restarting your computer, launch RStudio by searching for "RStudio" in the search bar





STEP 4: INSTALL AND LAUNCH POE TOOL

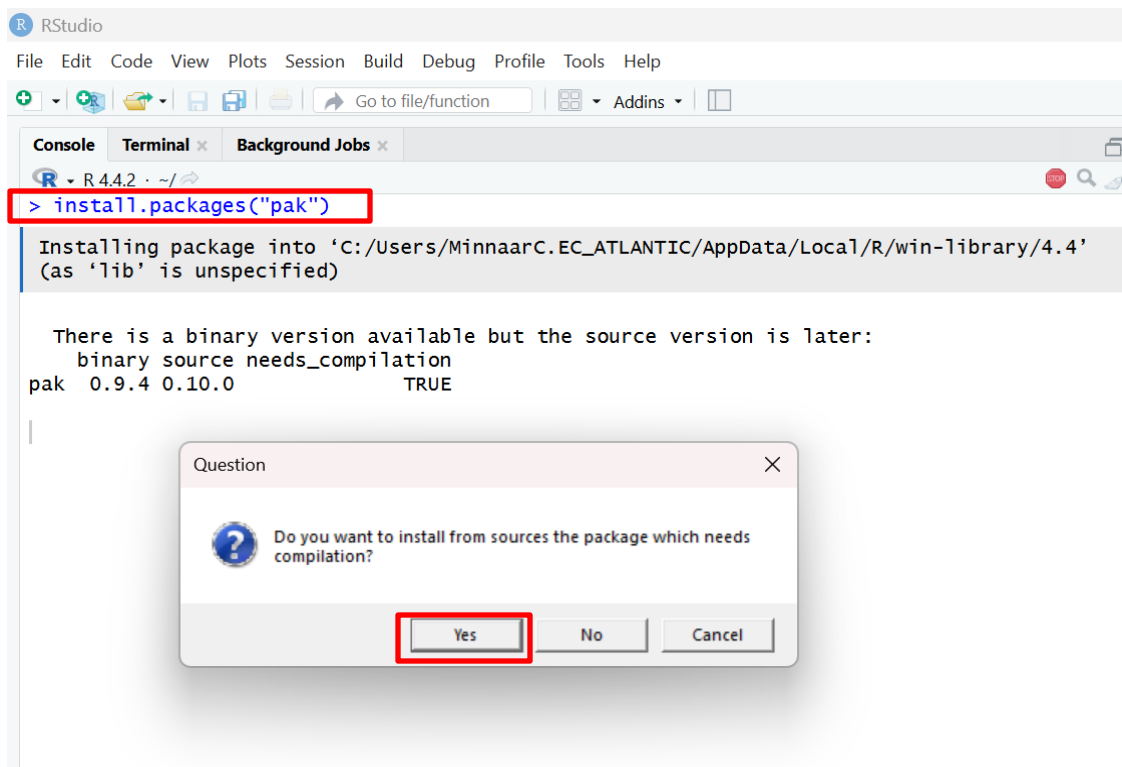
- **4.2** – Copy and paste the following text into your R Console on the left, then press the Enter key:

```
install.packages("pak")
```

If prompted with
"Do you want to install
from sources the packages
which needs compilation?"

Click **"Yes"**

Then, wait for the package
to install.





STEP 4: INSTALL AND LAUNCH POE TOOL

- **4.3** – When you see the blue ">" appear again, paste the following code into the console, then press Enter:

pak::pak("1catherinem/CWS-Pathway-of-Effects")

```
> pak::pak("1catherinem/CWS-Pathway-of-Effects")
```

```
✓ Updated metadata database: 4.98 MB in 3 files.
```

```
✓ Updating metadata database ... done
```

```
i No downloads are needed
```

```
✓ 1 pkg + 80 deps: kept 67 [34.2s]
```

```
> |
```

(Note: output text may differ from image example above, this is normal)





STEP 4: INSTALL AND LAUNCH POE TOOL

- **4.4** – Wait for the package to install. You'll know its done when you see the blue ">" appear again
- **If you did not receive any errors** after running this code, proceed to [Step 4.8.](#)
- **If you received an error** after running this code, then proceed to Steps 4.6 and 4.7 for troubleshooting.
- NOTE: If the PoE Shiny App is updated, this code must be run again to install the newest version.



STEP 4: INSTALL AND LAUNCH POE TOOL

- **4.6** – If you have not received any errors after running `pak::pak("1catherinem/CWS-Pathway-of-Effects")`, please skip this step.
- If an error is received like the one below and you have already installed RTools, then restart your computer.
- If that does not work, you will need to uninstall your current version of RTools and reinstall it, verifying you are installing the correct version for your version of R. **Proceed to Step 4.7.**

```
Error:
! error in pak subprocess
Caused by error:
! Could not find tools necessary to compile a package
Call `pkgbuild::check_build_tools(debug = TRUE)` to diagnose the problem.
i See `$stderr` for standard error.
Type .Last.error to see the more details.
> |
```

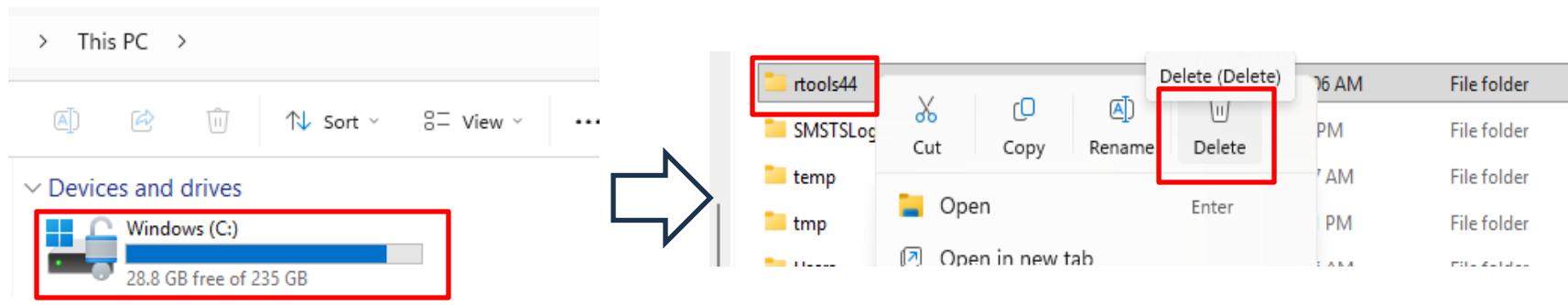


STEP 4: INSTALL AND LAUNCH POE TOOL

- **4.7** — If you have not received any errors after running `pak::pak("1catherinem/CWS-Pathway-of-Effects")`, please skip this step.

To uninstall RTools:

1. Open the Windows File Explorer.
2. Under "This PC", double-click on "Windows (C:)" in the left pane (may need to scroll) or in the main pane.
3. Find the RTools folder (may be called rtools44 or rtools45) and delete it to uninstall RTools.
4. Now, reinstall RTools by following the directions in [Step 2](#). After reinstalling RTools, re-attempt [Step 4](#).





STEP 4: INSTALL AND LAUNCH POE TOOL

- **4.8** – Once the package has successfully installed, you see the blue ">" appear again, paste the following code into the console:
`poe::poe_tool()`
- The PoE Shiny App tool will launch in your browser. If an RStudio window opens, click "Open in browser" at the top of the Window

Canadian Wildlife Service Pathways of Effects Assessment Tool (PEAT) Interactive View English Français

Build Pathways <

Valued Component ^

Sector ^

Select all Activities/Components that apply:

Mitigation Measures ^

Activities must be selected before mitigation measures

☐ Add Custom Mitigations





STEP 5: **QUICK LAUNCH** (AFTER FIRST INSTALLATION)

After installing the pak and poe packages for the first time, the tool is able to be launched with just a single line:

1. Open RStudio.
2. Type `poe::poe_tool()` in the console and press Enter.
3. The PoE Shiny App Tool will launch in your browser.





STEP 6: USE THE SHINY APP

- **6.1 – Welcome to the PoE Shiny App!**

The screenshot shows a web browser window displaying the 'Canadian Wildlife Service Pathways of Effects Assessment Tool (PEAT)' in 'Interactive View'. The interface is in English. On the left, there is a sidebar with sections: 'Build Pathways', 'Valued Component' (with a dropdown menu), 'Sector' (with a dropdown menu), 'Select all Activities/Components that apply:', 'Mitigation Measures' (with a note 'Activities must be selected before mitigation measures' and a radio button for 'Add Custom Mitigations'), and 'Report' (with input fields for 'Project name', 'Project status', and 'Notes...'). At the bottom of the sidebar is a 'Download report' button. The main area on the right is currently empty. A legend at the bottom right indicates: a blue square for 'Valued Component', a yellow square for 'Stressors/pressures associated with Activities/Components selected by user', and a red square for 'Potential Contravention of Legislation and/or Regulations'.



Note: You must keep R running while you use the Shiny App. The app was built for use in MS Edge or Google Chrome.



STEP 6: USE THE SHINY APP

- **6.2** – Select your preferred language (when available**)



****Note:** French not yet available as of July 2026 (awaiting translation). French will be made available as an option once all pathways have been added and finalized.



STEP 6: USE THE SHINY APP

- **6.3** – Select ONE Valued Component (VC) of interest from the Inputs dropdown menu**. This will build a PoE specific to that VC.

Canadian Wildlife Service Pathways of Effects Assessment Tool (PEAT) [Interactive View](#)

Build Pathways

Valued Component

Marine Birds

Migratory Birds and Bird SAR

Plant SAR

Terrestrial and Semi-Aquatic SAR

Wetlands

Test

Activities must be selected before mitigation measures

☐ Add Custom Mitigations

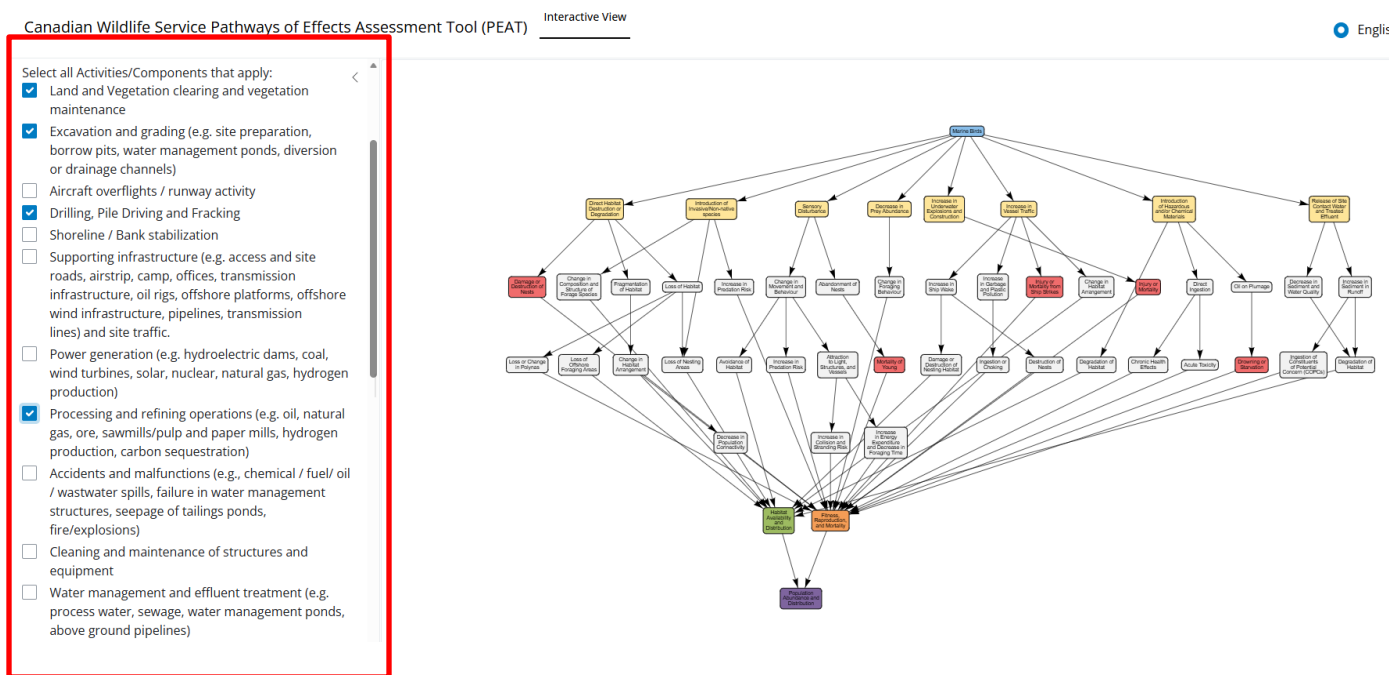


****Note:** For most project assessments, practitioners will need to build PoE diagrams for several (or all five) VCs listed, since projects can impact several VCs. In this case, choose a VC to start with



- **6.4** – Select ALL Activity/Components that apply to your project from the Inputs dropdown menu. The flowchart will automatically update with any new selections that are made.

- Note: you can scroll down to select additional Activities/Components





STEP 6: USE THE SHINY APP

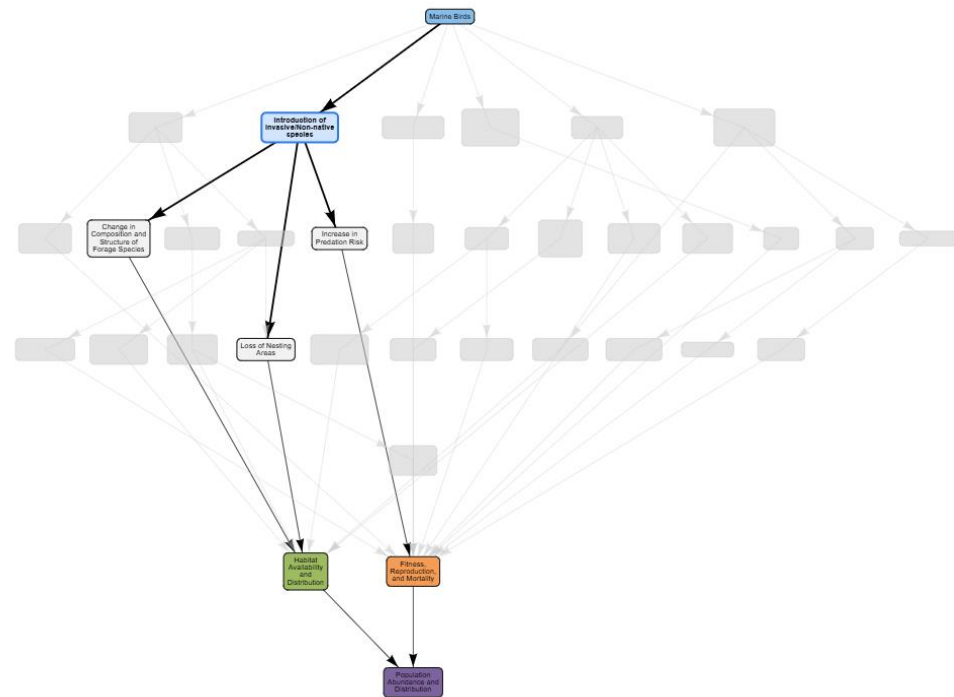
- **6.5** – The second dropdown contains "sectors" that pre-select a number of activities/components relevant to that sector.
- Additional activities/components can be selected to create a more complete PoE flowchart that better captures the potential impacts of a particular project





STEP 6: USE THE SHINY APP

- **6.6** – Click on any node in the flowchart to "focus" on its pathway of effect and grey out all other nodes.
- Click on any blank space bring the other nodes back into focus.





STEP 6: USE THE SHINY APP

- **6.7** – Mitigation measures are available in the left-hand panel and can be applied by clicking the checkboxes.
- Mitigated pathways will be greyed out in the current flowchart.

Canadian Wildlife Service Pathways of Effects Assessment Tool (PEAT) Interactive View En

☐ Vessels / shipping

Mitigation Measures

Select all Mitigations that apply:

- ☒ Avoid
- ☒ Minimize
- ☒ Restore On-site
- ☐ Offset

☐ Add Custom Mitigations

Report

Project name

Project status

Notes...

[Download report](#)





STEP 6: USE THE SHINY APP

- **6.8** – A report containing the current flowchart and applied mitigations can be generated and exported by entering the project information, and status and clicking "Download Report"
- NOTE: Reports are only able to be created and exported if the PoE Shiny App was launched using RStudio.

Report ^

Project name

Project status

Notes...

 Download report

Errors in the download can arise if the VPN drops the connection. If this happens, try again. If the error persists, refresh the page





CLOSING THE APP

- To close the program, simply close the MS Edge tab containing the tool and then close R.
- You may need to click the red "STOP" button to terminate the program before R is able to close.



FEEDBACK

- We welcome any feedback on the Shiny App or this guide!
- Please submit feedback to Peter Rodger
(Peter.Rodger@ec.gc.ca)